# Experiment 3: Where the Map Is Wrong

Insight analysis for Chapter 4, Section 4.3

Bolton Athit Davies

Reporting cutoff: 26 October 2026; tiny-map tables regenerated 6 November 2026

**Purpose of this document.** The raw write-up of Experiment 3 (`insights/experiment_3/`) reports that `map/map_tiny.db` places 96.6 % of its occupied cells on real structure and reproduces 89.5 % of the surfaces it observed. This document asks what the remaining 3.4 % and 10.5 % are, and whether the map's quality follows anything a reader would propose: range from the route, how often a surface was observed, the angle it was seen at, the loop-closure correction, the kind of object. It also opened, on 22 October, the four unregistered databases that then sat beside the scored one; those four were deleted on 6 November 2026 with the dataset decision that keeps only `map_tiny` of the older maps, so the tables below now carry the scored map alone and the second-map and provenance findings of 22 October are kept in prose as retired results, marked as such. Every number is written by `script/exp4_insights.py` into `output/compare/simulation/exp4-insights/`; the tables are input from there unchanged (regenerated 6 November on `map_tiny` alone, every `map_tiny` value identical to 22 October). One warehouse and one map support descriptive patterns, not statistics, and the document says so where it matters. Added 26 October 2026: the same classification applied to the two realistic-warehouse maps of the raw write-up (Insight 6), written by the script's `-realistic` entry into files named `*_realistic.*` beside the tiny ones, which it does not rewrite.

## 1 Questions and method

Q1. What are the wrong occupied cells: surfaces drawn too thick, or obstacles where there is nothing?
Q2. What are the missed surfaces, and does the miss follow range, the number of observing nodes, the viewing angle, or the object?
Q3. Did the loop closures change the map's accuracy?
Q4. On an independent route, where does the map block the robot, and why?
Q5. What were the four unregistered databases, and which recording built the scored map?
Q6. Do the two realistic-warehouse maps, four times the area and built without loop closure, fail by the same mechanism, and what blocks the independent probe on them?

**Raster.** The conventions are those of `script/validate_map.py`: the exported occupancy grid is placed in the world by the map pose that `script/fit_map_pose.py` measures from the grid alone, true structure is the world file's obstacle footprints filled plus the wall ring at 0.05 m per cell, true *surfaces* are the one-cell boundary of that, and a cell is correct within 0.25 m. Occupied cells beyond the tolerance are split at 0.50 m: nearer is a surface drawn too thick, further is a phantom. Recall is scored on the surfaces the map observed, as the chapter does.

**Per-node observation.** RTAB-Map stores, for every graph node, the local occupancy cells it contributed (obstacle, ground and empty, in the node frame). Moving them through the optimised node poses gives, for every world cell, how many nodes observed it and which node observed it first. The incidence angle of

a surface is the angle between its outward normal and the line of sight from the nearest graph node: 0° faced head-on, 90° seen edge-on along its own plane.

**Second map (retired).** On 22 October `map/map_02.db`, one of the unregistered databases, fitted the tiny world as well as the scored map (Q5), had been built on a different route, and was scored by the same method as the independent drivability probe that the lost `dataset_static_tiny_repeat` reference was meant to provide. The database was deleted on 6 November 2026; its 22 October scores are quoted below as retired results and appear in no table.

## 2 Insight 1: the wrong cells are thick surfaces, not phantom obstacles

Of the 3.4 % of occupied cells that miss the tolerance, nine in ten lie within 0.50 m of a real surface, and the rest form 22 clusters of at most ten cells. Table I.1 gives the split and figure I.1 the places.

**Table I.1:** Occupied map cells by their distance to the nearest true structure (filled shelf footprints and the wall ring of the world file). "Correct" is within the 0.25 m tolerance of the chapter; "thickened" is a cell beyond the tolerance but within 0.50 m of a true surface, i.e. a surface drawn too thick; "phantom" is further than that from any structure. The last columns describe the phantom cells: how many connected clusters they form, the size of the largest, and the median distance of a phantom cell from the recorded route.

| Map | occupied cells | correct [%] | thickened [%] | phantom [%] | clusters | largest [cells] | route dist. med. [m] |
|---|---|---|---|---|---|---|---|
| `map_tiny` | 11632 | 96.6 | 3.0 | 0.4 | 22 | 10 | 3.18 |

The thickened cells sit on the wall ring and on shelf fronts, where the stereo depth of a planar surface spreads over two or three cells. The 43 phantom cells together cover about 0.1 $m^2$; the largest cluster, ten cells at the north wall near the spawn, is 0.9 m from the route and was observed by one node. None of the phantoms lies in an aisle the recorded route uses, which is why the build route is fully drivable on this map.

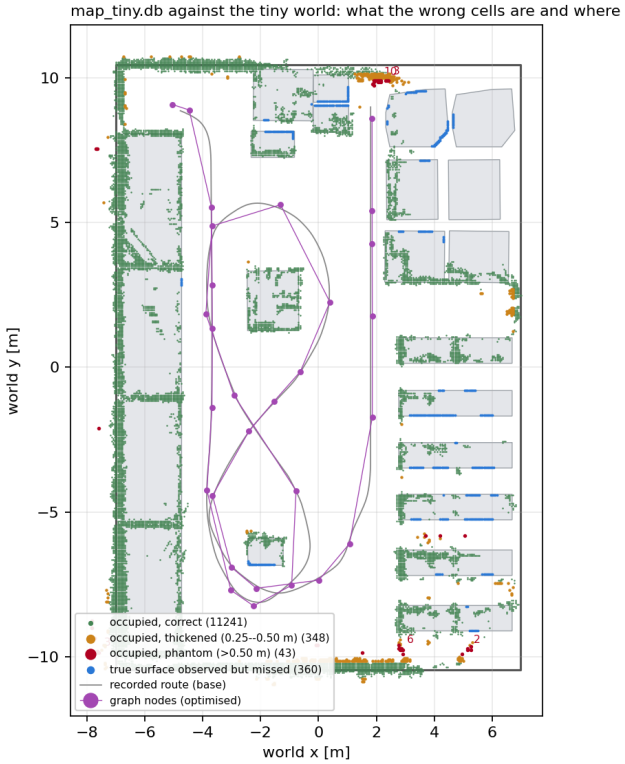



**Figure I.1:** `map_tiny.db` against the tiny world: correct occupied cells (green), surfaces drawn too thick (orange), phantom cells (red, the largest clusters labelled with their cell count), true surfaces the map observed but did not reproduce (blue), the recorded route and the optimised graph nodes.

# 3 Insight 2: surfaces are missed when they are seen edge-on and only once, not when they are far

Recall of the observed surfaces falls from 95 % for surfaces faced head-on to 76 % for surfaces seen at more than 60° incidence, and from 97 % for cells observed by three or more nodes to 78 % for cells observed by one; it does not fall with distance from the route within the 6 m the map reaches. Tables I.2 to I.4 and figure I.2 carry the three contrasts.

**Table I.2:** Recall of the observed surfaces by the incidence angle of the nearest graph node's line of sight on the surface: 0° is a surface faced head-on, 90° one seen edge-on along its own plane (the faces of neighbouring shelves across a narrow gap, looked into from the aisle). Normals come from the world footprints; the viewing node is the nearest optimised graph node.

| Incidence angle [°] | `map_tiny` surface cells | `map_tiny` recall [%] |
|---|---|---|
| 0–30 | 1453 | 95.3 |
| 30–60 | 1120 | 92.4 |
| 60–90 | 861 | 76.1 |

**Table I.3:** Precision and recall of the observed surfaces by the number of graph nodes whose stored local grid covers the cell. A map that improves with repeated observation would show both rising with the count.

| Observing nodes | `map_tiny` precision cells | [%] | `map_tiny` recall cells | [%] |
|---|---|---|---|---|
| 1 | 2529 | 91.2 | 834 | 78.2 |
| 2 | 3631 | 96.8 | 854 | 90.5 |
| 3–4 | 4642 | 99.4 | 1246 | 97.4 |
| 5–7 | 661 | 99.7 | 238 | 97.9 |
| ≥ 8 | 1 | 100.0 | 0 | — |

**Table I.4:** Precision and recall of the observed surfaces by distance from the recorded route (`dataset_map_tiny`, the route that built `map_tiny.db`; for `map_02.db` the same route is only a proxy for its own, unknown route). Stereo depth error grows with range, so a range effect would show as falling precision and recall with distance.

| Distance from route [m] | `map_tiny` precision cells | [%] | `map_tiny` recall cells | [%] |
|---|---|---|---|---|
| 0–1 | 819 | 97.3 | 436 | 89.0 |
| 1–2 | 3211 | 96.4 | 932 | 88.7 |
| 2–3 | 3389 | 97.4 | 836 | 86.0 |
| 3–4 | 3183 | 97.6 | 862 | 94.3 |
| 4–6 | 1015 | 91.1 | 364 | 88.7 |
| 6–8 | 15 | 100.0 | 4 | 100.0 |
| ≥ 8 | 0 | — | 0 | — |

The two mechanisms are the same surfaces seen from the same place. The missed cells in figure I.1 are the faces of neighbouring shelves across the narrow gaps on the east side, which the route looks into from the aisle but never enters: each face is seen once, from a node in the aisle, at grazing incidence. Precision tells the same story from the other side: cells observed by one node are 91 % correct, cells observed by three or more 99 %. Stereo range error, the mechanism a reader would propose first, does not show: precision stays at 96–98 % out to 4 m and recall has no trend with distance.

Table I.5 lists the same effect by object. The objects with the lowest recall were each observed by a single node, while the one large shelf observed by four nodes is reproduced at 99 %. Which objects these are follows from the route, not from the objects: the figure-eight and the one straight along the east wall decide what is faced and what is grazed.

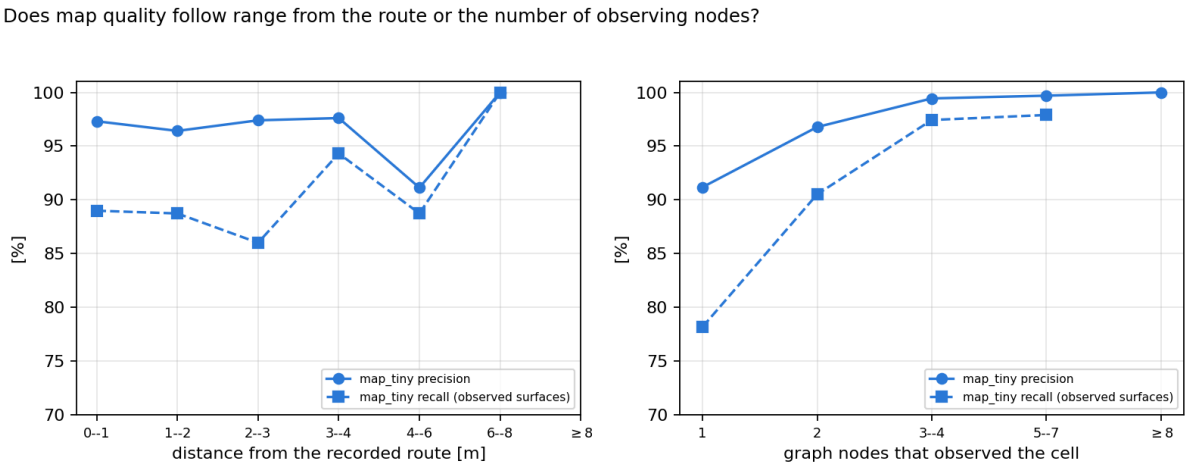



**Figure I.2:** Precision of the occupied cells and recall of the observed surfaces against distance from the recorded route (left) and against the number of graph nodes that observed the cell (right), for both tiny-warehouse maps.

**Table I.5:** Recall of `map_tiny.db` on the observed surfaces of each world object, worst first: surface cells the map observed, their share of the object's surface, the share of them that the map reproduces within tolerance, the median distance of those cells from the recorded route, and the median number of graph nodes that observed them.

| Object | observed [cells] | of surface [%] | recall [%] | route dist. med. [m] | nodes med. |
|---|---|---|---|---|---|
| ClutteringA_01_016 | 10 | 6 | 0.0 | 2.9 | 1 |
| ShelfD_01_001 | 103 | 55 | 48.5 | 1.9 | 1 |
| ClutteringA_01_017 | 107 | 66 | 52.3 | 1.1 | 1 |
| Bucket_01_020 | 68 | 100 | 52.9 | 1.5 | 1 |
| ShelfE_01_002 | 159 | 85 | 69.2 | 2.8 | 1 |
| ShelfD_01_002 | 128 | 68 | 69.5 | 2.3 | 1 |
| Bucket_01_022 | 82 | 100 | 72.0 | 1.4 | 2 |
| TrashCanC_01_002 | 90 | 100 | 72.2 | 1.8 | 3 |
| Bucket_01_021 | 84 | 100 | 75.0 | 0.9 | 2 |
| ClutteringC_01_027 | 114 | 75 | 84.2 | 1.1 | 3 |
| ShelfD_01_003 | 148 | 80 | 91.2 | 3.2 | 2 |
| ClutteringC_01_030 | 82 | 55 | 91.5 | 0.6 | 3 |
| ShelfE_01_003 | 156 | 83 | 94.2 | 4.1 | 1 |
| ClutteringC_01_032 | 138 | 100 | 96.4 | 2.3 | 2 |
| ShelfF_01_001 | 802 | 100 | 99.4 | 2.4 | 4 |
| ShelfE_01_001 | 169 | 90 | 100.0 | 2.7 | 2 |
| ClutteringC_01_028 | 67 | 44 | 100.0 | 3.6 | 1 |
| ClutteringC_01_031 | 150 | 100 | 100.0 | 1.2 | 4 |
| wall ring | 777 | 56 | 100.0 | 3.2 | 2 |

# 4 Insight 3: the loop closures corrected the graph, and the map did not need it

The five closure links corrected the odometry by at most 0.26 m in the plane and 0.52 m in height, and the occupied cells first observed after the closing revisit are as accurate as those observed before it. Table I.6 and figure I.3 carry the numbers.

**Table I.6:** Loop closures of `map_tiny.db` and the correction the graph optimisation applied: planar and vertical distance between each node's odometry pose and its optimised pose, by node range, and the precision of the occupied cells first observed before and after the revisit that closed the loop (nodes 1–18 against 19–29).

| Quantity | nodes 1–18 | nodes 19–29 | all |
|---|---|---|---|
| planar correction, median [m] | 0.01 | 0.16 | 0.09 |
| planar correction, max [m] | 0.12 | 0.26 | 0.26 |
| vertical correction, max [m] | 0.18 | 0.52 | 0.52 |
| occupied cells first seen | 10098 | 1366 | 11632 |
| precision of those cells [%] | 96.9 | 95.6 | 96.6 |
| closure links (from → to): 9→22, 10→23, 22→9, 23→10, 30→29 | | | |

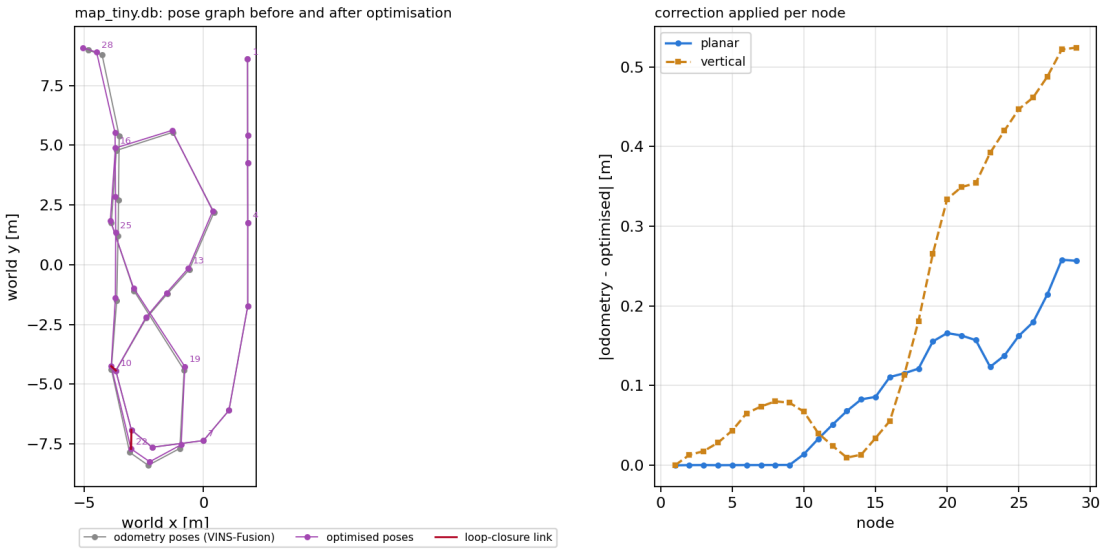



**Figure I.3:** `map_tiny.db`: odometry poses and optimised poses of the 29 nodes in the optimised graph, with the loop-closure links (left), and the planar and vertical correction each node received (right).

The correction is real but small, and it is mostly vertical: VINS-Fusion's odometry climbed 0.5 m over the 70 m route, and the closures at nodes 22 and 23, which revisit nodes 9 and 10, pulled the second half of the graph back onto the floor. In the plane the correction stays under the 0.25 m tolerance for every node, so the cells placed from corrected and from uncorrected poses score alike. On a route of this length the map's accuracy was set by the odometry, with the closures as insurance that was not drawn on.

# 5 Insight 4: on an independent route the second map blocks the robot at its phantom cells

**retired**: the database that gave this measurement was deleted on 6 November 2026 and the result cannot be regenerated. For the record, on 22 October `map_02.db` under the true footprint along

the recorded route blocked 7.6 % of the poses and left 5.5 % unmapped, in three stretches (poses 125–164, 922–986 and 1485–1539, with a short 2356–2421 stretch at 0.10 m clearance), every one on a phantom cluster of that map. Tables I.7 and I.8 now carry the scored map alone, for which the build route is clear by construction; the 22 October tables and the second map's cell figure are kept under `trash/output/exp4-insights-tiny-with-map_02_20261106/`.

**Table I.7:** Drivability of both tiny-warehouse maps under the nav2 footprint stamped along the recorded route of `dataset_map_tiny` (3032 poses). For `map_tiny.db` this is the route that built the map; for `map_02.db` it is an independent route through the same world, which the lost `dataset_static_tiny_repeat` reference was meant to provide.

| Map | route | false obstacle [% of poses] | unmapped [%] | clearance med. [m] | clearance p05 [m] |
|---|---|---|---|---|---|
| `map_tiny` | built the map | 0.0 | 0.0 | 1.01 | 0.46 |

**Table I.8:** Contiguous stretches of the recorded `dataset_map_tiny` route on which the true footprint overlaps an occupied cell of the map: pose range, number of poses, location, and the smallest clearance from the route centre to a mapped obstacle inside the stretch.

| Map | poses | count | $x$ [m] | $y$ [m] | min. clearance [m] |
|---|---|---|---|---|---|
| `map_tiny` | — | 0 | — | — | — |

This was the measurement the raw write-up could not make. The build route is drivable on the map it built by construction, since the camera cleared the cells the robot passed through; an independent route meets the cells the build route only looked at. On the second map the three blocked stretches sit on its phantom clusters of 3, 16 and 3 cells, in the figure-eight and at the start of the east straight; the route centre comes within 0.0 m to 0.2 m of them. Phantoms that cover 0.1 $m^2$ of a 290 $m^2$ floor are negligible for geometry and decisive for drivability when they fall in an aisle.

**[Placement of the second map, retired]** `map_02.db` was placed by its own grid fit (agreement 0.385, margin 54 % over the best rival yaw), not by a known spawn, so its per-cell precision carried that uncertainty. With the database deleted the placement cannot be re-examined; the retired numbers above are quoted with that caveat and no further.

# 6 Insight 5: the scored map was built from the recorded tiny bag, and the large-warehouse attempt left a database

The nodes of `map_tiny.db` coincide with the ground truth of `dataset_map_tiny` at the same stamps to a median 0.05 m, so the map was built from that recording and the write-up's self-check is a true self-check (the identity test is regenerated: median 0.05 m, maximum 0.10 m); and `map_03.db` was a 76 m run whose poses climbed to 9.8 m above the floor, the large-warehouse attempt that the chapter reports as having left no artefact. Table I.9 now describes the scored database alone; the four others were described on 22 October and deleted on 6 November 2026, so what follows about them is retired.

`map_02.db` (38 nodes, 55.3 MB) fitted the tiny world as well as the scored map and its nodes did not follow the recorded route (median 4.1 m from the ground truth at the same stamps): a second map of the same world on a different route, made on 21 September, and the only candidate for the lost `dataset_static_tiny_repeat` session. `map_03.db` (18 nodes, 29.2 MB) could not be placed in any large world (agreement 0.19, margin 7 %); its nodes ran 37 m along one aisle and left the floor, which is tracking failure, not mapping failure. `map_01.db` and `map_02_vins.db` were two-node starts. All four are deleted.

**Table I.9:** The five RTAB-Map databases under `map/`: size, graph nodes, simulation-time span of the nodes, odometry path length and planar extent, vertical range of the node poses, loop closures, the placement fit against the tiny world (agreement score and margin over the best rival yaw, from `script/fit_map_pose.py`), and what the database is identified as. Only `map_tiny.db` is registered in the project records.

| Database | MB | nodes | stamps [s] | path [m] | extent [m] | $z$ [m] | closures | fit (agr., margin) | identified as |
|---|---|---|---|---|---|---|---|---|---|
| `map_tiny.db` | 67.4 | 30 | 31–90 | 69.5 | 17×7 | -0.0..0.6 | 5 | 0.387, +61 % | the scored tiny map (registered); built on the `dataset_map_tiny` route |

**[Provenance]** The identifications of `map_02.db` and `map_03.db` rested on their contents alone; no record names them, and since 6 November the databases cannot be re-examined, so the large-warehouse attempt is again without an artefact. The raw write-up's statement that the scored map's node stamps match no recorded dataset is contradicted by the stamp-and-pose match above and is withdrawn.

# 7 Insight 6: on the realistic maps the shelves are thickened and the walls displaced; the October builds are blocked at cells laid during initialisation, the November I build at cells laid in the aisles

Of the occupied cells of the four realistic maps, 9.7 to 15.4 % are surfaces drawn too thick and 5.2 to 7.3 % are phantoms, thirteen to eighteen times the tiny map's phantom share; every probe pose the maps block lies within 0.6 m of a phantom cell, and in three of the four maps 63 to 96 % of those blocking cells were first observed by one of the first five graph nodes — in the fourth, the November I build, 18 % were, and the blocked stretches are mid-route. Table I.10 gives the split, figures I.4 to I.7 the places, and tables I.14 and I.15 the blocked stretches. The four maps are two builds of each of two mapping recordings; the builds differ in three back-end parameters at once (`Rtabmap/DetectionRate` 4 against 1 Hz, `Grid/RangeMax` 8 against 10 m, `Grid/MaxGroundHeight` 0.29 against 0.10 m; October against November), so a difference between builds is read for its signature, not attributed.

**Table I.10:** Occupied map cells by their distance to the nearest true structure (filled shelf footprints and the wall ring of the world file). "Correct" is within the 0.25 m tolerance of the chapter; "thickened" is a cell beyond the tolerance but within 0.50 m of a true surface, i.e. a surface drawn too thick; "phantom" is further than that from any structure. The last columns describe the phantom cells: how many connected clusters they form, the size of the largest, and the median distance of a phantom cell from the independent probe route.

| Map | occupied cells | correct [%] | thickened [%] | phantom [%] | clusters | largest [cells] | route dist. med. [m] |
|---|---|---|---|---|---|---|---|
| `U-Oct` | 26711 | 81.3 | 13.4 | 5.2 | 413 | 108 | 4.89 |
| `I-Oct` | 25395 | 85.0 | 9.7 | 5.3 | 385 | 157 | 10.10 |
| `U-Nov` | 22701 | 77.4 | 15.4 | 7.3 | 440 | 91 | 4.91 |
| `I-Nov` | 23220 | 82.2 | 11.5 | 6.3 | 537 | 108 | 7.91 |

The thickening is the tiny map's mechanism at the shelves, and the phantoms are a different one at the walls, in all four maps. The shelf rows the build route passed are green with an orange rim, surfaces spread over two or three cells as on the tiny map; the phantoms form a red band the full length of the east wall about 0.6 m inside its true line, and a second along the south wall outside it. The largest clusters of the October maps, 108 and 70 cells on U and 157 and 117 on I, all sit on that east band, 6 to 11 m from the probe route and first observed at median node 32 and 36; the November maps add clusters on the south and west walls, which their longer grid range reaches. The walls are the structure furthest from any build node; what moved them is not identified here.

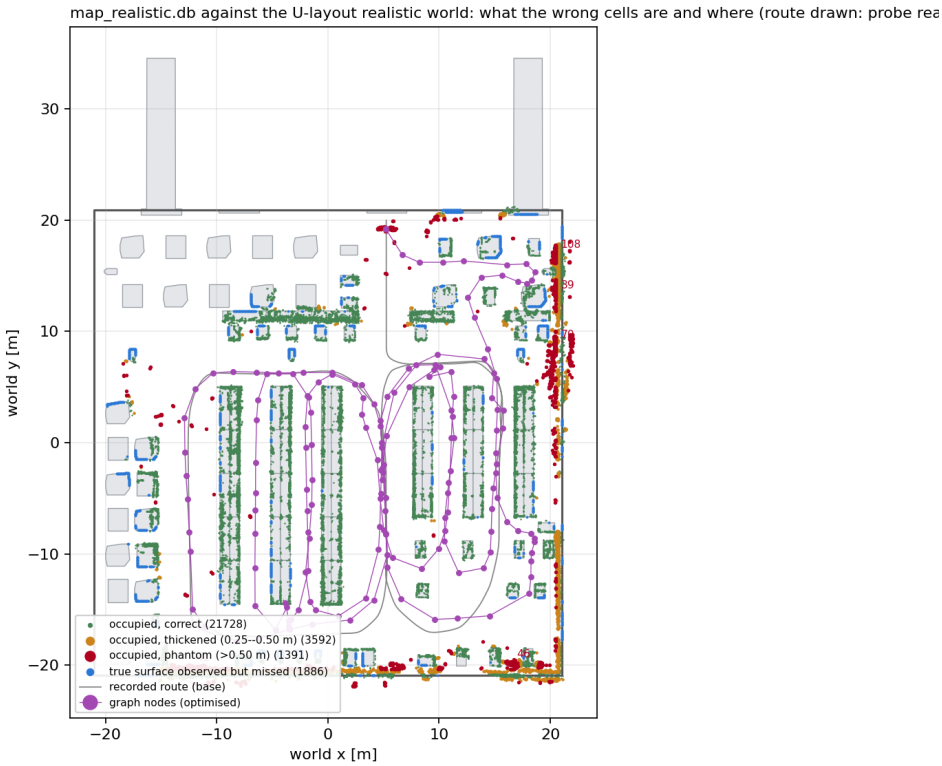



**Figure I.4:** `map_realistic.db` (U, October) against the U-layout realistic world, placed by its fit; conventions as figure I.1. The route drawn is the independent probe `realistic_000`, which did not build the map; the graph nodes are the build route. Phantom clusters are labelled with their cell count.

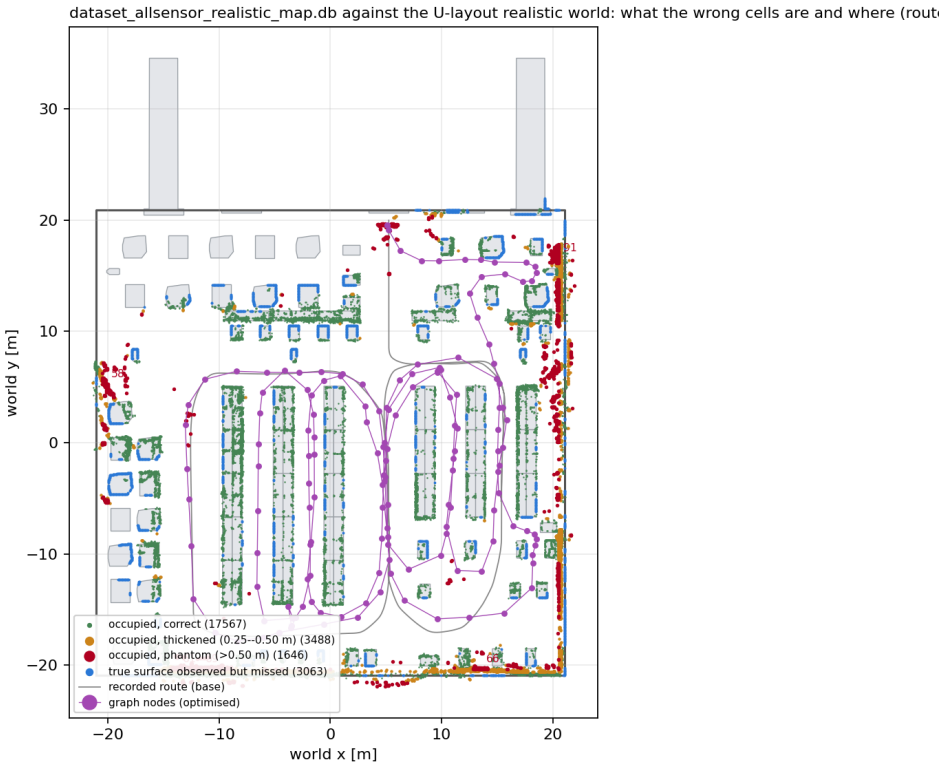



**Figure I.5:** `dataset_allsensor_realistic_map.db` (U, November) against the same world and probe. The 10 m grid range reaches the west wall and the outer bays, which carry new phantom clusters; the spawn stretch is longer than in October.

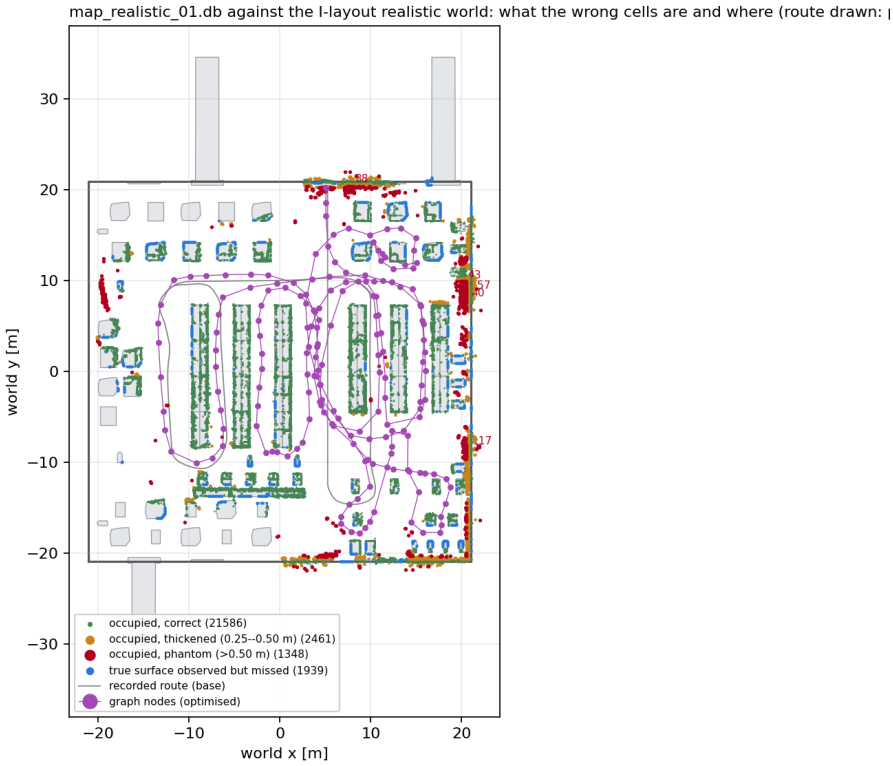



**Figure I.6:** `map_realistic_01.db` (I, October) against the I-layout realistic world; the route drawn is the probe `realistic_001`.

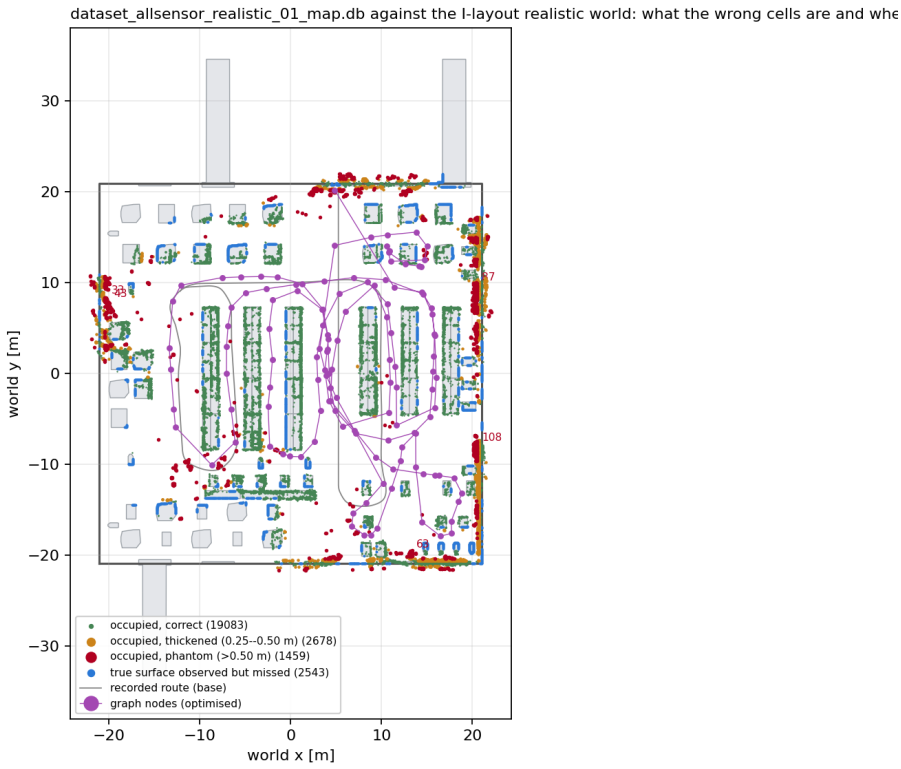



**Figure I.7:** `dataset_allsensor_realistic_01_map.db` (I, November) against the same world and probe. The blocked stretches sit in the aisles of the probe's left loop and along its bottom straight, on phantom cells first observed mid-run.

**Table I.11:** Precision and recall of the observed surfaces by the number of graph nodes whose stored local grid covers the cell, for the two realistic-warehouse maps (U is `map_realistic`, I is `map_realistic_01`). The tiny-map contrast was 91 to 99 percent precision and 78 to 97 percent recall from one node to three or more.

| Observing nodes | U-Oct precision cells | [%] | U-Oct recall cells | [%] | I-Oct precision cells | [%] | I-Oct recall cells | [%] | U-Nov precision cells | [%] | U-Nov recall cells | [%] | I-Nov precision cells | [%] | I-Nov recall cells | [%] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4891 | 65.5 | 1884 | 71.8 | 4068 | 83.2 | 1788 | 75.8 | 4045 | 69.1 | 2259 | 55.5 | 4052 | 75.4 | 2063 | 61.5 |
| 2 | 6048 | 75.3 | 1828 | 81.8 | 6236 | 80.3 | 2153 | 79.8 | 6672 | 73.5 | 1781 | 81.0 | 6107 | 85.8 | 1941 | 83.6 |
| 3–4 | 7958 | 82.0 | 2967 | 86.1 | 8006 | 82.9 | 3017 | 82.4 | 5035 | 74.7 | 2349 | 82.0 | 7026 | 79.2 | 3161 | 84.4 |
| 5–7 | 5261 | 98.9 | 2708 | 90.1 | 4992 | 93.2 | 2296 | 85.3 | 4675 | 82.6 | 3133 | 77.8 | 3862 | 86.8 | 2430 | 80.7 |
| $\geq 8$ | 1274 | 100.0 | 840 | 91.0 | 1315 | 99.9 | 732 | 87.7 | 2157 | 100.0 | 2183 | 83.0 | 1412 | 100.0 | 771 | 84.7 |

**Table I.12:** Recall of the observed surfaces by the incidence angle of the nearest graph node's line of sight on the surface: 0° is a surface faced head-on, 90° one seen edge-on along its own plane (the faces of neighbouring shelves across a narrow gap, looked into from the aisle). Normals come from the world footprints; the viewing node is the nearest optimised graph node.

| Incidence angle [°] | U-Oct surface cells | U-Oct recall [%] | I-Oct surface cells | I-Oct recall [%] | U-Nov surface cells | U-Nov recall [%] | I-Nov surface cells | I-Nov recall [%] |
|---|---|---|---|---|---|---|---|---|
| 0–30 | 6042 | 85.4 | 5445 | 82.6 | 6668 | 79.1 | 5480 | 78.9 |
| 30–60 | 2413 | 84.1 | 2535 | 82.1 | 2592 | 74.6 | 2993 | 77.2 |
| 60–90 | 2454 | 74.7 | 2477 | 78.4 | 2921 | 65.3 | 2728 | 74.2 |

**Table I.13:** Precision and recall of the observed surfaces by distance from the independent probe route (`realistic_000` for U, `map_realistic`; `realistic_001` for I, `map_realistic_01`). The probe is not the route that built either map, so this is only a proxy for range from the mapping camera.

| Distance from route [m] | U-Oct precision cells | [%] | U-Oct recall cells | [%] | I-Oct precision cells | [%] | I-Oct recall cells | [%] | U-Nov precision cells | [%] | U-Nov recall cells | [%] | I-Nov precision cells | [%] | I-Nov recall cells | [%] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0–1 | 110 | 6.4 | 1 | 100.0 | 1281 | 91.7 | 415 | 95.2 | 132 | 0.8 | 1 | 100.0 | 1145 | 87.1 | 424 | 91.5 |
| 1–2 | 2343 | 99.0 | 1142 | 82.4 | 2581 | 96.6 | 907 | 95.0 | 1670 | 99.4 | 1141 | 81.9 | 2600 | 94.6 | 911 | 93.9 |
| 2–3 | 4387 | 97.7 | 2248 | 88.5 | 5172 | 96.3 | 1875 | 90.7 | 3069 | 96.6 | 2272 | 81.2 | 4510 | 96.1 | 1855 | 90.4 |
| 3–4 | 6575 | 80.5 | 2130 | 87.1 | 2655 | 96.5 | 1200 | 80.4 | 6069 | 77.1 | 2205 | 82.4 | 1890 | 95.6 | 1181 | 66.8 |
| 4–6 | 8608 | 81.5 | 3369 | 82.4 | 4475 | 95.8 | 2188 | 78.0 | 7492 | 77.9 | 3902 | 77.9 | 3326 | 95.2 | 2107 | 69.7 |
| 6–8 | 2202 | 64.9 | 1182 | 68.9 | 3073 | 84.5 | 1497 | 80.4 | 2086 | 64.9 | 1656 | 49.2 | 3784 | 79.5 | 1936 | 80.6 |
| $\geq 8$ | 2486 | 55.3 | 837 | 77.1 | 6158 | 56.7 | 2375 | 70.9 | 2183 | 48.9 | 1004 | 66.0 | 5965 | 55.5 | 2787 | 68.9 |

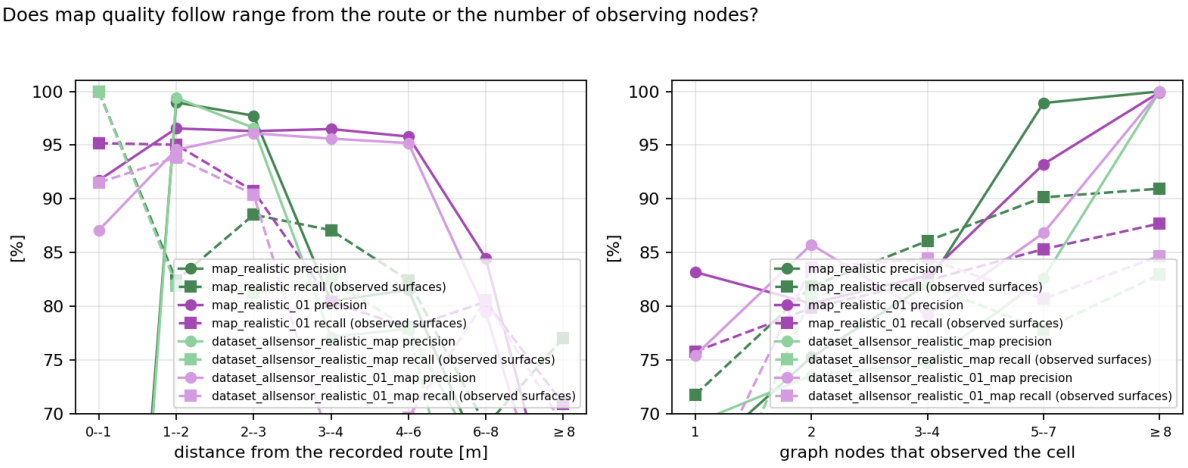



**Figure I.8:** Precision of the occupied cells and recall of the observed surfaces against distance from the independent probe route (left, a proxy only: the probe did not build any map) and against the number of graph nodes that observed the cell (right), for the four realistic-warehouse maps.

The view-count mechanism of Insight 2 holds in the October builds and is larger: cells observed by one node are 66 and 83 % correct, cells observed by five or more 99 and 100 %, against 91 and 99 % on the tiny map. In the November builds the curve is flatter below five observations (69 to 87 %) and the 1 Hz detection rate gives every cell fewer observations, which is the direction the mechanism predicts for that parameter. The incidence effect holds in all four, 79 to 85 % head-on against 65 to 78 % grazing, but the head-on figure is ten to sixteen points below the tiny map's 95 %, so viewing geometry explains the spread between surfaces and not the level they all sit at.

**Table I.14:** Drivability of the two realistic-warehouse maps under the nav2 footprint stamped along the static-twin recording of each layout, which did not build the map (`realistic_000`, 6278 poses, for the U map; `realistic_001`, 4847 poses, for the I map). Same method as the raw write-up, which it reproduces.

| Map | route | false obstacle [% of poses] | unmapped [%] | clearance med. [m] | clearance p05 [m] |
|---|---|---|---|---|---|
| `U-Oct` | independent | 2.1 | 2.4 | 1.95 | 0.61 |
| `I-Oct` | independent | 4.4 | 0.7 | 2.16 | 0.36 |
| `U-Nov` | independent | 5.0 | 1.9 | 1.99 | 0.36 |
| `I-Nov` | independent | 10.7 | 0.5 | 1.97 | 0.18 |

**Table I.15:** Contiguous stretches of the independent probe route of each layout on which the true footprint overlaps an occupied cell of the map: pose range, number of poses, location, and the smallest clearance from the route centre to a mapped obstacle inside the stretch.

| Map | poses | count | $x$ [m] | $y$ [m] | min. clearance [m] |
|---|---|---|---|---|---|
| `U-Oct` | 108–184 | 77 | 5.2 | 19.5 | 0.00 |
| `U-Oct` | 3332–3360 | 29 | -12.5 | 1.7 | 0.21 |
| `U-Oct` | 284–311 | 28 | 5.2 | 15.3 | 0.00 |
| `I-Oct` | 0–149 | 150 | 5.2 | 20.0 | 0.20 |
| `I-Oct` | 1836–1868 | 33 | 10.0 | -3.2 | 0.00 |
| `I-Oct` | 3431–3459 | 29 | -12.2 | -3.6 | 0.15 |
| `I-Oct` | 3729–3729 | 1 | -9.2 | -10.6 | 0.40 |
| `U-Nov` | 0–223 | 224 | 5.2 | 19.5 | 0.05 |
| `U-Nov` | 284–313 | 30 | 5.2 | 15.3 | 0.00 |
| `U-Nov` | 3299–3327 | 29 | -12.4 | 2.7 | 0.10 |
| `U-Nov` | 3360–3385 | 26 | -12.6 | 0.9 | 0.05 |
| `U-Nov` | 3395–3399 | 5 | -12.6 | 0.2 | 0.42 |
| `I-Nov` | 0–141 | 142 | 5.2 | 20.0 | 0.16 |
| `I-Nov` | 3901–3968 | 68 | -5.9 | -6.4 | 0.00 |
| `I-Nov` | 1326–1380 | 55 | 6.6 | -14.4 | 0.00 |
| `I-Nov` | 3629–3678 | 50 | -11.3 | -9.8 | 0.00 |
| `I-Nov` | 3753–3792 | 40 | -7.8 | -10.5 | 0.16 |
| `I-Nov` | 3845–3872 | 28 | -6.1 | -8.6 | 0.05 |
| `I-Nov` | 1909–1935 | 27 | 9.9 | -1.1 | 0.10 |
| `I-Nov` | 1971–1997 | 27 | 9.9 | 0.7 | 0.20 |

The probes are blocked by the phantoms, not by the thickened shelving, and in the October builds the blocking phantoms are the oldest cells in each map. Of 134 and 213 blocked poses, 134 and 212 lie within 0.6 m of a phantom cell and none within that distance of a thickened cell alone; the largest stretch on each map is the first metres of the probe at the spawn, 77 and 150 poses; and the 77 and 41 phantom cells under the blocked stretches have a median first-observing node of 1, with 96 and 63 % first seen by node 5 or earlier against 6 and 3 % of all phantom cells. The raw write-up records that node 1 of each database is not at identity and that its height settles over the first nodes, the inertial initialisation transient; the cells laid during it are the ones under the robot's starting position, where every probe begins.

The November I build is blocked by a different population. Its 518 blocked poses (10.7 %) include 27

within 0.6 m of thickened cells only, the first such on any map, and the 79 phantom cells under its stretches have a median first-observing node of 27 with only 18 % early; the stretches of 68, 55 and 50 poses sit in the aisles of the probe's left loop and on its bottom straight. A ground-height threshold lowered from 0.29 to 0.10 m admits the lowest sensed cells in the aisles as obstacles, which is this signature; the detection rate and grid range changed in the same build and are not excluded. The November U build keeps the October pattern (224 spawn poses, 88 % of its blocking cells early), so the aisle population is specific to the I layout's narrower aisles, untested.

**[Two rasters, three agreements and one disagreement]** The chapter's `validate_map.py` reports 3.8, 5.0, 4.4 and 10.2 % blocked poses; this document's raster, whose origin is the wall ring rather than the union of map, route and wall, reports 2.1, 5.0, 4.4 and 10.7 %. The one disagreement is the October U spawn stretch, where the chapter counts probe poses 0–183 and this raster 108–184 at 0.00 m clearance, so the count flips on a sub-cell offset. The chapter's figures stand as the registered ones.

**[Wall displacement, untested]** The east band is inside the wall and the south band outside it in all four maps, which a uniform scale error would not produce; drift accumulated along the closure-free build route is the candidate, but no test separates it from a range-dependent stereo bias at the 3 to 5 m the walls were seen from. The build-route ground truth on the rescued drive would decide it.

**[Parameter attribution, untested]** Each of the three parameters that changed between the builds has a signature consistent with the November drop, and none is isolated: a rebuild changing one parameter at a time is the test, and the front-end session of the November builds, made during the latency runs of Experiment 3, is not recorded.

## 8 What these data cannot answer

- **Whether the patterns generalise.** One map of one small world; the contrasts are descriptive. The view-count and incidence effects agreed with the second map on 22 October and with how stereo mapping works, which is the reason to trust them, not a test.
- **The old large-warehouse attempt.** `map_03.db` existed until 6 November and could not be scored: no placement fitted, and the pose graph left the floor; it is deleted. The four realistic-warehouse maps of Insight 6 now stand as the large result; whether their patterns hold under loop closure is untested, because they were built without it, and which of three parameters produced the November drop is untested, because they changed together.
- **Whether the second map's phantoms were typical.** Their cause was not identified; they appeared where that map's own route started and turned, which suggested transient depth at initialisation, the mechanism Insight 6 later found on the realistic maps; the database is deleted.
- **Distance beyond 6 m.** The map reaches no further, so the absence of a range effect is bounded to that range.

## 9 Summary

**Consequences for Chapter 4.** Report the map's error as surface thickness plus a handful of phantom cells, and the recall gap as a viewing-geometry effect of the route, not a sensor-range effect. The independent-route drivability of 22 October can no longer be quoted: its database is deleted, and the chapter reports the tiny map's drivability reference as lost. Correct the statement that the scored map matches no recorded dataset. Report that a database identified from its contents as the large-warehouse attempt existed until 6 November and was deleted unplaced and unscored, so the attempt is again without an artefact and its failure stands as tracking loss on the progress record. For the realistic maps, attribute the false obstacles to phantom cells laid during initialisation and at the walls, not to thickened shelving, and state that discarding the cells of the first few nodes, or starting the map after the inertial transient has

**Table I.16:** Questions, answers and the strength of the evidence.

| | Answer | Evidence | Confidence |
|---|---|---|---|
| Q1 wrong cells | Thick surfaces, not phantoms: 348 thickened against 43 phantom cells in 22 small clusters | table I.1, figure I.1 | high |
| Q2 missed surfaces | Seen edge-on and once: recall 95 % head-on to 76 % grazing, 97 % at three nodes to 78 % at one; no range effect within 6 m | tables I.2 to I.4 | medium, one map |
| Q3 loop closures | Corrected 0.26 m planar and 0.52 m vertical; map accuracy unchanged across the revisit | table I.6 | medium |
| Q4 independent route | Retired: on 22 Oct the second map blocked 7.6 % of poses, all at its phantom clusters; its database is deleted | prose of Insight 4 | retired |
| Q5 provenance | Scored map built from `dataset_map_tiny` (regenerated); `map_02` was a second tiny map and `map_03` the large attempt, both deleted 6 Nov | table I.9, prose | high / retired |
| Q6 realistic maps | Shelves thickened as on the tiny map; walls displaced about 0.5 m as phantom bands in all four; October builds blocked only at spawn-transient phantoms (96 % / 63 % laid by node 5), the November I build mid-route in the aisles (18 %); three parameters changed together | tables I.10 and I.15 | medium, one map per cell |

settled, is the one correction the data predict would remove the spawn stretch, 57 to 71 % of the blocked poses on three of the four maps. Report the November builds as the current configuration scored lower on every metric, with the drop unattributed until one parameter is changed at a time.